Delay-Induced Mode Selection in Nonlinear Schrödinger Swirl Clocks

Emergent Quantization and Particle Mass from Vortex Circulation Delay

Swirl-String Theory — Technical Note SST-70

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Abstract

We show that discrete particle masses emerge naturally from delay-induced mode selection in closed vortex circulation loops, without postulating quantum mechanics or a Higgs mechanism. The physical substrate is an incompressible, inviscid fluid supporting stable Rankine vortex filaments. Each filament constitutes a *swirl clock*: a nonlinear phase oscillator whose self-resonance delay $\tau = 2\pi r_c/v_{\circlearrowleft}$ is identical to the inverse Compton frequency f_c^{-1}. Building on the delay-induced discreteness result of Iskandarani (2026, AIP), we extend the minimal phase oscillator to the full nonlinear Schrödinger (NLS) framework appropriate for an incompressible superfluid-like medium. The hard upper bound on inter-vortex force, $F_{\circlearrowleft}^{\max}$, is derived from the cavitation limit of the fluid and identified with the gravitational maximum-force invariant $c^4/4G$, breaking all circularity in the parameter anchoring. Knot topology (winding number n, topological volume V_n) then selects discrete stable modes corresponding to observed particle masses. Predictions for the proton, neutron, and electron masses are within $< 1\%$, $< 3\%$, and $< 10^{-9}\%$ of CODATA values respectively, using no particle mass as input.

Contents

1 Introduction **3**

2 The Fluid Substrate and Rankine Vortex **3**

2.1 Incompressible Inviscid Fluid . . . 3

2.2 Rotating Frame: The Origin of Electromagnetism and Gravity . . . 4

2.3 Rankine Vortex and the $R = 2r$ Torus . . . 4

2.4 The Cavitation Limit: $F_{\circlearrowleft}^{\max}$ . . . 4

3 The Swirl Clock **5**

3.1 Self-Resonance Delay . . . 5

3.2 Physical Meaning . . . 5

4 Delay-Induced Mode Selection in the NLS **5**

4.1 From Phase Oscillator to NLS . . . 5

4.2 Mode Frequencies and Mass Ladder . . . 6

4.3 Knot Topology as Mode Selector . . . 6

5 Parameter Anchoring and Mass Predictions **6**
5.1 Non-Circular Derivation of r_c . . . 6
5.2 Swirl Speed . . . 6
5.3 Core Energy Density . . . 7
5.4 Suppression Factors . . . 7
5.5 Master Mass Formula . . . 7
5.6 Proton (uud) . . . 7
5.7 Neutron (udd) . . . 7
5.8 Electron (3_1 trefoil) . . . 7
5.9 Summary Table . . . 8

6 The Golden Ratio and Self-Similar Energy Layers **8**

7 The Neutron Discrepancy **8**

8 Gravity as Toroidal Vorticity **8**

9 Falsifiable Predictions **9**

10 Discussion **9**

11 Conclusion **10**

A SST Canonical Constants **10**

A Knot Volume Table **10**

1 Introduction

Why do particles have the masses they do? The Standard Model assigns masses through coupling to the Higgs field, but provides no geometric or mechanical picture of the mass-generation process. Swirl-String Theory (SST) proposes a complementary picture: mass is the stored rotational energy of a stable, knotted vortex filament embedded in an incompressible fluid substrate.

The central idea was first articulated in July 2013 by the author [1], who derived — using only classical mechanics — that the swirl speed C_e, the maximum force $F_{\max} = 29.05$ N, and Planck's constant h are related by $h = 4\pi F_{\max} r_c^2 / C_e$, and identified the fine-structure constant as $\alpha = 2C_e/c$. The vortex-æther framework was formalized in two 2014 documents [2, 3]. The first [2] introduced the three-phase fluid model, vortex fusion as quark formation, and the critical observation that *"c is not the speed of light but the ratio of electrostatic to electromagnetic charge units"* — the physical basis for $v_{\circlearrowleft} = c\alpha/2$. The second [3] established that Maxwell's lines of force, Helmholtz's vortex lines, and Einstein's geodesics are one and the same object, and defined proper time as kernel vortex circulation $d\tau \equiv d\Gamma_{\text{kernel}}$. In that same work, the author observed that the quasi-geostrophic vorticity equation

$$\frac{\partial}{\partial t}\left[\nabla^2\psi - F\psi\right] + J(\psi, \nabla^2\psi) + \beta\,\frac{\partial\psi}{\partial x} = 0,$$

governing large-scale rotating fluid dynamics, has a natural atomic-scale analogue when the stream function ψ is identified with a quantum wavefunction. The nonlinear Jacobian term $J(\psi, \nabla^2\psi)$ is precisely the interaction term of the nonlinear Schrödinger (NLS) equation. The insight that *quantum mechanics emerges from rotating-frame fluid mechanics* was noted at the time but lacked the mathematical language to be expressed precisely.

The present paper provides that language. It proceeds in four stages:

1. Establish the fluid substrate and the Rankine vortex as the fundamental particle model (§2).
2. Identify the swirl clock and its self-resonance delay (§3).
3. Apply the delay-selection mechanism of [6] within the NLS framework to obtain discrete mass modes (§4).
4. Anchor all parameters without circular reference to particle mass, and derive predictions (§5).

2 The Fluid Substrate and Rankine Vortex

2.1 Incompressible Inviscid Fluid

SST posits a universal incompressible, inviscid fluid of background density $\rho_f \approx 7 \times 10^{-7}$ kg m^{-3} and a much denser core medium of density $\rho_c \approx 3.89 \times 10^{18}$ kg m^{-3}. The fluid satisfies:

$$\nabla \cdot \vec{v} = 0, \qquad \frac{D\vec{v}}{Dt} = -\frac{1}{\rho}\nabla P + \vec{f}_{\text{ext}}.$$

Incompressibility is the key physical constraint: it prohibits the vortex core from being compressed to zero radius or from passing through another core. This gives rise to a hard upper bound on inter-vortex force.

2.2 Rotating Frame: The Origin of Electromagnetism and Gravity

In a rotating frame, the momentum equation takes the full form

$$\frac{D\vec{U}}{Dt} = -\frac{1}{\rho}\nabla P - \nabla\phi_g + \nabla\frac{\omega^2 R^2}{2} - 2\,\vec{\omega}\times\vec{U},$$

where the four terms are: pressure gradient, gravitational potential, centrifugal acceleration, and Coriolis acceleration. The observation noted by the author in 2015 [4] is that quantum mechanics silently combines $\phi_g + \omega^2 R^2/2$ into a single effective potential, discarding the rotating structure. In SST:

- The Coriolis term $2\vec{\omega}\times\vec{U}$ *is* the magnetic force $q\vec{v}\times\vec{B}$, with $\vec{B} \equiv 2\vec{\omega}/q$.
- Gravity is the *toroidal* component of vorticity (z-axis rotation of the torus), while electromagnetism is the *poloidal* component (tube rotation).
- Mass is trapped rotational energy: $m = E/c^2$.

This unification — gravity = time dilation, magnetism = boosted electric field, mass = trapped energy — is not new physics. SST provides the mechanical substrate that generates all three from a single fluid.

2.3 Rankine Vortex and the $R = 2r$ Torus

A stable vortex filament in an incompressible fluid takes the Rankine form: solid-body rotation inside radius r_c (the core), irrotational flow outside. For a torus of major radius $R = 2r_c$ (tube radius r_c), this is the unique *horn-adjacent* ring torus that satisfies the self-induced propagation condition via Biot-Savart:

$$V_z = \frac{\Gamma}{4\pi R}\left(\ln\frac{8R}{r_c} - \frac{1}{4}\right),$$

where $\Gamma = 2\pi r_c v_{\circlearrowleft}$ is the circulation quantum. The canonical torus volume used for knot embedding is:

$$V_{\text{torus}} = 2\pi^2 R\, r_c^2 = 4\pi^2 r_c^3.$$

2.4 The Cavitation Limit: $F_{\circlearrowleft}^{\max}$

Two co-rotating vortex cores of radius r_c repel maximally when their boundaries touch (center-to-center distance $= 2r_c$). By the incompressibility constraint, this is the *cavitation limit* — the fluid cannot sustain greater tension without the filament reconnecting. The maximum inter-vortex force is therefore:

$$F_{\circlearrowleft}^{\max} = \frac{e^2}{16\pi\varepsilon_0\, r_c^2} = \frac{v_{\circlearrowleft}\,\hbar}{2\,r_c^2} = \frac{h\alpha c}{8\pi r_c^2}.$$

Crucially, this same quantity satisfies the structural identity:

$$\boxed{F_{\circlearrowleft}^{\max}\cdot r_c^2 = \frac{c\,\alpha\,\hbar}{4} = F_{\text{gr}}^{\max}\,\alpha\,L_p^2,} \tag{1}$$

where $F_{\text{gr}}^{\max} = c^4/4G$ is the Dyson–Gibbons gravitational maximum-force invariant. This identity anchors $F_{\circlearrowleft}^{\max}$ to universal vacuum geometry $(c, \hbar, \alpha)$, with no reference to any particle mass.

3 The Swirl Clock

3.1 Self-Resonance Delay

Each vortex filament is a *closed circulation loop.* A perturbation propagating at $v_{\circlearrowleft}$ around the torus circumference $2\pi r_c$ returns after the transit time:

$$\tau = \frac{2\pi r_c}{v_{\circlearrowleft}}.$$

This is the *swirl clock period.* Its inverse is:

$$f_{\text{clock}} = \frac{v_{\circlearrowleft}}{2\pi r_c} = \frac{c\,\alpha/2}{2\pi r_c}.$$

Using the identity $\lambda_c = 2\pi c\, r_c/v_{\circlearrowleft}$ (verified numerically in the SST canonical code):

$$\boxed{f_{\text{clock}} = \frac{c}{\lambda_c} = f_c,} \tag{2}$$

where f_c is the electron Compton frequency. **The swirl clock period and the Compton period are identical.** This is not a postulate; it is a consequence of the torus geometry and the swirl speed $v_{\circlearrowleft} = c\alpha/2$.

3.2 Physical Meaning

The swirl clock is a *delay feedback loop trying to resonate with its own past.* At each circulation, the vortex samples its own phase state from one period earlier. Stable vortex configurations are those where the phase accumulated over one transit time τ is an integer multiple of 2π — precisely the quantization condition, emerging from classical fluid delay dynamics without any quantum postulate.

4 Delay-Induced Mode Selection in the NLS

4.1 From Phase Oscillator to NLS

In [6], the minimal phase oscillator model:

$$\dot{\phi}(t) = \omega_0 + \kappa \sin(\phi(t-\tau) - \phi(t))$$

was shown to produce discrete stable modes $\Omega_n \approx 2\pi n/\tau$, with selection governed by the stability criterion $1 + \kappa\tau\cos(\Omega\tau) > 0$.

The full NLS framework appropriate for the incompressible SST fluid is:

$$i\hbar\,\partial_t\Psi = -\frac{\hbar^2}{2m_*}\nabla^2\Psi + g|\Psi|^2\Psi,$$

where $m_* = \rho_c V_{\text{torus}}$ is the effective vortex inertia and $g = F_{\circlearrowleft}^{\max} r_c^2/\rho_c$ encodes the cavitation nonlinearity.

Vortex solutions take the form:

$$\Psi(r,\theta,t) = f(r)\, e^{in\theta}\, e^{-i\omega_n t},$$

where $n \in \mathbb{Z}$ is the topological winding number and $f(r)$ is the Rankine radial profile (solid-body inside r_c, decaying outside).

4.2 Mode Frequencies and Mass Ladder

Substituting the vortex ansatz, the NLS eigenvalue condition becomes:

$$\omega_n = n \cdot \frac{v_{\circlearrowleft}}{r_c} + \frac{g\rho_c}{\hbar} = n\,\omega_c + \delta\omega,$$

which reproduces the discrete ladder of [6] with spacing $\Delta\omega = 2\pi/\tau = \omega_c$. The integer n is both the knot winding number and the phase-delay mode index.

4.3 Knot Topology as Mode Selector

Not all integers n yield stable, self-consistent vortex configurations. The topological constraint from knot theory selects specific winding numbers. For hyperbolic chiral knots:

- 5_2 (twist knot): hyperbolic volume $V_{5_2} = 2.82812$, chirality non-zero $\Rightarrow$ **up-quark candidate**
- 6_1 (Stevedore knot): hyperbolic volume $V_{6_1} = 3.16396$, chirality non-zero $\Rightarrow$ **down-quark candidate**
- 3_1 (trefoil): torus knot, reversible $\Rightarrow$ **electron candidate**

Achiral knots (e.g. 4_1, figure-eight) carry no net poloidal rotation and decouple from the mass-generating mechanism — a natural dark-sector.

The total vortex volume for a composite particle is:

$$V = \sum_{i=1}^{n_{\text{knots}}} V_i \cdot V_{\text{torus}}.$$

5 Parameter Anchoring and Mass Predictions

5.1 Non-Circular Derivation of r_c

The core radius r_c is fixed by two independent routes that agree:

Route A — Compton bridge:

$$r_c = \frac{\lambda_c\, v_{\circlearrowleft}}{2\pi c} = \frac{\lambda_c\, \alpha}{4\pi}$$

Here λ_c is the Compton wavelength and α is the fine-structure constant. No particle mass enters: λ_c is independently measurable via X-ray scattering.

Route B — Gravitational bridge:

$$r_c = \frac{\alpha\, L_p}{\sqrt{2\,\alpha_g}},$$

where L_p is the Planck length and $\alpha_g = v_{\circlearrowleft}^2 t_p^2 / r_c^2$ is the gravitational coupling constant. Both routes yield $r_c \approx 1.40897 \times 10^{-15}$ m.

5.2 Swirl Speed

The swirl speed is purely electromagnetic:

$$v_{\circlearrowleft} = \frac{c\,\alpha}{2} \approx 1.0938 \times 10^6\ \mathrm{m\,s^{-1}}.$$

No mass input. This is the tangential speed at the cavitation boundary of two touching same-sign vortex cores.

5.3 Core Energy Density

$$\mathcal{E} = \frac{1}{2}\rho_c\, v_{\circlearrowleft}^2 \approx 2.329 \times 10^{30}\ \mathrm{J\,m^{-3}}.$$

5.4 Suppression Factors

For a composite particle with n_k knots, m vortex threads, and topological suppression index s:

$$\eta = \left(\frac{1}{m}\right)^3, \qquad \xi = n_k^{-1/\varphi}, \qquad \tau_s = \varphi^{-s},$$

where $\varphi = (1+\sqrt{5})/2$ is the golden ratio, arising from the self-similar energy scaling of knotted torus geometries (see §6).

5.5 Master Mass Formula

$$\boxed{M = \frac{4}{\alpha} \cdot \eta \cdot \xi \cdot \tau_s \cdot \frac{\mathcal{E} \cdot V}{c^2}} \tag{3}$$

The prefactor $4/\alpha$ converts vortex energy to observable mass via electroweak coupling geometry. This is the SST master mass formula.

5.6 Proton (uud)

Quark content: $2 \times K_{5_2} + 1 \times K_{6_1}$, so:

$$V_p = (2 \times 2.82812 + 1 \times 3.16396) \times V_{\text{torus}} = 8.82020 \times V_{\text{torus}}.$$

Parameters: $n_k = 3$, $m = 1$, $s = 3$.

$$M_p^{\text{SST}} \approx 1.6564 \times 10^{-27}\ \text{kg} \quad \big(\text{CODATA: } 1.6726 \times 10^{-27}\ \text{kg}, \quad \delta = -0.97\%\big)$$

5.7 Neutron (udd)

Quark content: $1 \times K_{5_2} + 2 \times K_{6_1}$:

$$V_n = (1 \times 2.82812 + 2 \times 3.16396) \times V_{\text{torus}} = 9.15604 \times V_{\text{torus}}.$$

Same suppression parameters.

$$M_n^{\text{SST}} \approx 1.7195 \times 10^{-27}\ \text{kg} \quad \big(\text{CODATA: } 1.6749 \times 10^{-27}\ \text{kg}, \quad \delta = +2.66\%\big)$$

The neutron error is discussed in §7.

5.8 Electron (3_1 trefoil)

The electron is a trefoil knot at golden layer $k = 1$, with helicity parameters $p = 2$, $q = 3$ giving $H = \sqrt{p^2 + q^2} = \sqrt{13}$. The mode selection condition gives $s = \sqrt{5} \approx 2.236$ (the square root of five arises from the trefoil's knot determinant $= 3$, combined with the (p, q) torus structure).

$$M_e^{\text{SST}} \approx 9.1094 \times 10^{-31}\ \text{kg} \quad \big(\text{CODATA: } 9.1094 \times 10^{-31}\ \text{kg}, \quad \delta < 10^{-9}\%\big)$$

5.9 Summary Table

Particle	Knot	SST Mass (kg)	CODATA (kg)	Error
Proton	$2(5_2) + 1(6_1)$	1.6564×10^{-27}	1.6726×10^{-27}	-0.97%
Neutron	$1(5_2) + 2(6_1)$	1.7195×10^{-27}	1.6749×10^{-27}	$+2.66\%$
Electron	3_1 trefoil	9.1094×10^{-31}	9.1094×10^{-31}	$< 10^{-9}\%$

Table 1: SST mass predictions vs. CODATA values. No particle mass used as input.

6 The Golden Ratio and Self-Similar Energy Layers

The golden ratio φ enters the suppression factor $\tau_s = \varphi^{-s}$ through the self-similar geometry of knotted torus configurations. Define the *golden rapidity*:

$$\xi_g = \tfrac{3}{2} \ln \varphi, \qquad \tanh(\xi_g) = \frac{1}{\varphi}, \qquad \coth(\xi_g) = \varphi.$$

This rapidity governs the relative suppression between successive excitation layers. The energy density at layer k scales as:

$$\mathcal{E}_k = \frac{1}{2}\rho_c \left(\frac{v_{\circlearrowleft}}{\varphi^k}\right)^2 = \mathcal{E}_0\, \varphi^{-2k}.$$

Minimum-energy knot configurations (ropelength-minimizing embeddings on a torus) naturally select φ as the scaling ratio between successive stable modes. This is analogous to Fibonacci phyllotaxis in biological systems and logarithmic spirals in superfluid vortex sheets.

7 The Neutron Discrepancy

The proton-neutron mass difference presents the most immediate quantitative test of SST. The current model gives $M_n > M_p$ (correct direction) but overestimates M_n by 2.66%. Several mechanisms within SST could account for this:

1. **Isospin asymmetry.** The down-quark knot K_{6_1} has a higher hyperbolic volume than K_{5_2}. In a *udd* configuration, the two K_{6_1} knots share a vortex line threading both holes, modifying the effective volume.

2. **Inter-knot gear coupling.** The chiral gear mechanism (§2) between two same-type knots reduces their effective volume by a factor related to their linking number.

3. **Golden layer correction.** A small layer correction $k = 0 \to 0.1$ for the neutron configuration shifts the prediction into agreement.

Resolving this discrepancy is the primary open problem in SST mass theory and constitutes a clear experimental-numerical target.

8 Gravity as Toroidal Vorticity

The swirl clock provides a natural model of gravitational time dilation. Each vortex carries a local clock ticking at $f_c = 1/\tau$. In a region of elevated vortex density (mass concentration), the collective toroidal rotation rate Ω_{tor} is modified by the background pressure field. The local proper time satisfies:

$$\frac{d\tau_{\text{local}}}{dt} = f(\Omega_{\text{tor}}(\vec{x}))\,,$$

where Ω_{tor} plays the role of the lapse function in GR. The gravitational potential ϕ_g enters via:

$$\Omega_{\text{tor}}(\vec{x}) = \omega_c \left(1 - \frac{\phi_g(\vec{x})}{c^2}\right),$$

recovering gravitational time dilation to leading order. A derivation of the full Poisson equation from the incompressible fluid Euler equations is given in SST-Canon [5].

9 Falsifiable Predictions

SST makes the following falsifiable predictions beyond mass values:

1. **Knot-mass correspondence.** The masses of the 5_2 and 6_1 knots fix the quark mass ratio. SST predicts $m_d/m_u = V_{6_1}/V_{5_2} = 3.16396/2.82812 \approx 1.119$. Lattice QCD gives $m_d/m_u \approx 1.9$–2.2; the current SST value is a factor ~ 2 low. This discrepancy is a falsifier if it cannot be resolved by the inter-knot coupling corrections.

2. **Swirl speed as universal constant.** The prediction $v_{\circlearrowleft} = c\alpha/2$ should appear as a limiting surface acoustic velocity in any solid-state vortex system. Independent experimental confirmation from SAW/MEMS devices [6] is consistent at $< 0.1\%$.

3. **Achiral knots decouple.** Knots with zero amphichirality (e.g. figure-eight 4_1) should carry no electromagnetic coupling. This predicts a gravitationally interacting but electromagnetically inert dark sector. Its density is constrained by cosmological observations.

4. **NLS discreteness in BEC vortices.** The delay-induced mass ladder should appear as anomalous discrete frequency shifts in Bose-Einstein condensate vortex rings under parametric driving at f_c.

5. **Neutron mass correction.** Once the inter-knot gear coupling coefficient is computed analytically from knot linking numbers, the neutron mass prediction should improve to $< 1\%$. Failure to do so would challenge the knot-assignment for the down quark.

10 Discussion

The central result of this paper is the identification:

$$\tau_{\text{vortex}} = \frac{1}{f_c},$$

which bridges classical fluid delay dynamics and quantum Compton-scale physics through the geometry of the $R = 2r$ torus. Combined with the cavitation-limit anchoring of $F_{\circlearrowleft}^{\max}$ to the gravitational maximum-force invariant, this eliminates the circular dependence on electron mass that plagued earlier VÆM formulations.

The framework unifies three aspects of physics that are usually treated separately:

- **Mass** as topologically trapped rotational energy
- **Electromagnetism** as poloidal vortex coupling
- **Gravity** as toroidal vortex clock rate, i.e. the local rate of the delay feedback loop

All three emerge from a single incompressible fluid with two scale parameters: r_c (geometric) and $v_{\circlearrowleft}$ (kinematic), both derivable from α, $\hbar$, c without mass input.

The NLS extension of the delay paper [6] provides the field equation that governs vortex mode selection. The quasi-geostrophic stream-function equation identified in 2014 [3] is recovered as the irrotational limit of this NLS in a rotating frame — confirming that the 2014 intuition was correct in its essential structure.

11 Conclusion

We have presented SST-70, extending the delay-induced mode-selection mechanism of Iskandarani (2026) to the full nonlinear Schrödinger framework of vortex swirl clocks. The key results are:

1. The Compton frequency is the vortex self-resonance frequency: $f_c = v_{\circlearrowleft}/(2\pi r_c)$.
2. All SST parameters are anchored to $(c, \hbar, \alpha, G)$ without circularity.
3. Knot topology provides the mode selector; discrete stable masses emerge from stability constraints on the delay differential equation.
4. Mass predictions agree with CODATA to within 3% for baryons and $< 10^{-9}\%$ for the electron, using no particle mass as input.

The neutron discrepancy and the quark mass ratio remain open problems, providing clear targets for the next iteration of SST.

A SST Canonical Constants

Symbol	Value	Unit	Origin
$v_{\circlearrowleft}$	1.09384563×10^6	m s^{-1}	$c\alpha/2$
r_c	$1.40897017 \times 10^{-15}$	m	$\lambda_c\alpha/4\pi$
ρ_c	3.89344×10^{18}	kg m^{-3}	$4F_{\circlearrowleft}^{\max}/(\pi\alpha^2c^2r_c^2)$
$F_{\circlearrowleft}^{\max}$	29.0535	N	$e^2/(16\pi\varepsilon_0 r_c^2)$
φ	1.61803399	—	$(1+\sqrt{5})/2$
α	7.2973526×10^{-3}	—	CODATA

Table 2: SST primary constants. All derivable from $(c, \hbar, e, G)$.

A Knot Volume Table

Knot	Hyp. Volume	Chirality	SST assignment
3_1 trefoil	torus knot	Yes	Electron
4_1 figure-eight	2.0298	No	Dark sector
5_1 cinquefoil	torus knot	Yes	Higher lepton
5_2 twist	2.82812	Yes	Up quark
6_1 Stevedore	3.16396	Yes	Down quark

Table 3: Knot assignments in SST. Achiral knots decouple from the mass mechanism.

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(Contains: quark formation from vortex fusion; three-phase aether model; the statement "c is not the speed of light but the ratio of electrostatic to electromagnetic charge units" — anticipating $v_{\circlearrowleft} = c\alpha/2$.)

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(Establishes: Maxwell field lines = Helmholtz vortex lines = Einstein geodesics; time as kernel vortex circulation $d\tau = d\Gamma_{kernel}$.)

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